

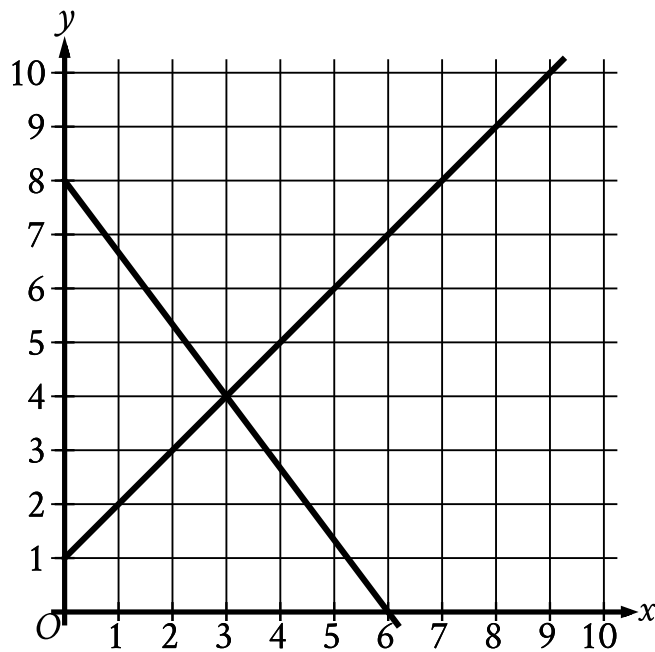
☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

16



- For the first line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - (0 comma 8)
 - (3 comma 4)
 - (5 comma four thirds)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following points:
 - (0 comma 1)
 - (3 comma 4)
 - (5 comma 6)

The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

A. $(2, 3)$

B. 3,4

C. (4,5)

D. (5,6)

Last week, an interior designer earned a total of \$ 1,258 from consulting for x hours and drawing up plans for y hours. The equation $68x + 85y = 1,258$ represents this situation. Which of the following is the best interpretation of 68 in this context?

- A. The interior designer earned \$ 68 per hour consulting last week.
- B. The interior designer worked 68 hours drawing up plans last week.
- C. The interior designer earned \$ 68 per hour drawing up plans last week.
- D. The interior designer worked 68 hours consulting last week.

Q3

GRID-IN ALGEBRA

$$2.6 + x = 2.8$$

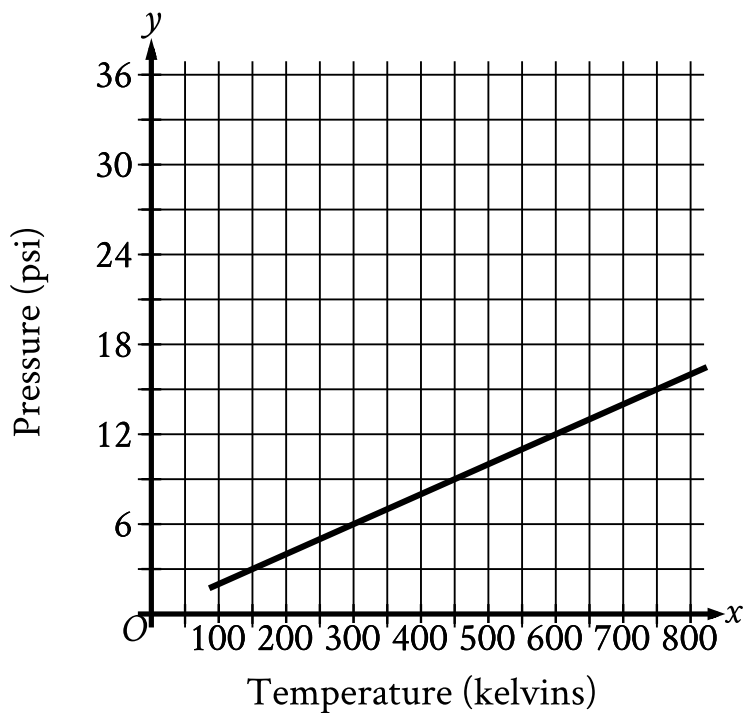
What value of x is the solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Argon is placed inside a container with a constant volume. The graph shows the estimated pressure y , in pounds per square inch psi, of the argon when its temperature is x kelvins.



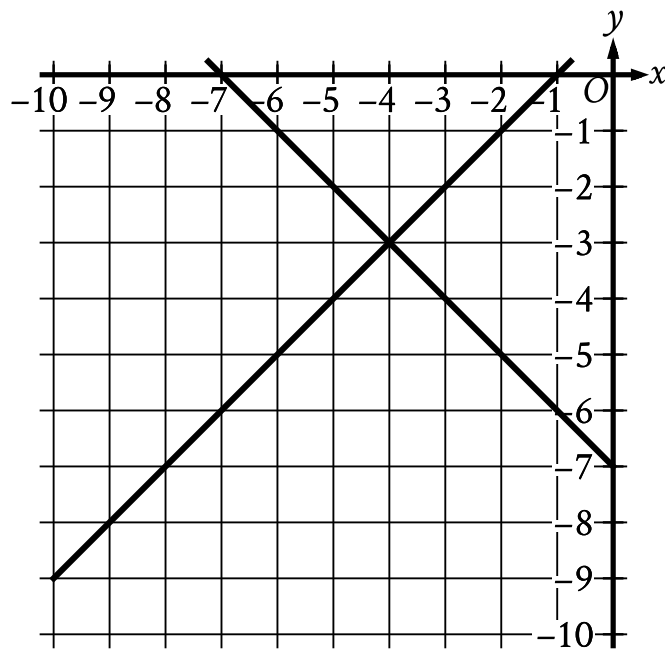
- The line slants gradually up from left to right.
- The line begins at the approximate point (90 comma 1.8).
- The line passes through the following points:
 - (300 comma 6)
 - (450 comma 9)
 - (600 comma 12)

What is the estimated pressure of the argon, in psi, when the temperature is 600 kelvins?

- A. 6
- B. 12

C. 300

D. 600



- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (negative 7 comma 0)
 - (negative 4 comma negative 3)
 - (0 comma negative 7)
- For the second line in the system:
 - The line slants gradually up from left to right.
 - The line passes through the following points:
 - (negative 5 comma negative 4)
 - (negative 4 comma negative 3)
 - (negative 3 comma negative 2)

The graph of a system of linear equations is shown. What is the solution x, y to the system?

A. 0, -7

B. 0, -3

C. -4, -3

D. -4, 0

A shipment consists of 5-pound boxes and 10-pound boxes with a total weight of 220 pounds. There are 13 10-pound boxes in the shipment. How many 5-pound boxes are in the shipment?

- A. 5
- B. 10
- C. 13
- D. 18

If $3x = 30$, what is the value of $3x - 12$?

A. -2

B. 18

C. 22

D. 42

$$f(x) = 7x + 1$$

The function gives the total number of people on a company retreat with x managers. What is the total number of people on a company retreat with 7 managers?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN ALGEBRA

$$x = 8$$

$$x + 3y = 26$$

The solution to the given system of equations is x, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

For a camping trip a group bought x one-liter bottles of water and y three-liter bottles of water, for a total of 240 liters of water. Which equation represents this situation?

- A. $x + 3y = 240$
- B. $x + y = 240$
- C. $3x + 3y = 240$
- D. $3x + y = 240$